

PGR106

Linear Algebra

Lecture 4 – System of Linear Equations



Linear Equation

- The equation

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

- which expresses the real or complex quantity ***b*** in terms of the unknowns $x_1, x_2, \dots, x_n$ and the real or complex constants $a_1, a_2, \dots, a_n$ is called a **linear equation**.
- A system of ***m*** linear equations in ***n*** unknowns, $x_1, x_2, \dots, x_n$, or a **linear system**, is a set of ***m*** linear equations each in ***n*** unknowns.
- In linear algebra, data is represented in the form of **linear equations**.
- In data science, linear equations **can be** represented as **matrices** and **vectors**.

Linear Equation

- Linear equation is an equation of the form:

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b,$$

$a_1, a_2, a_3, \dots, a_n$ are coefficients

$x_1, x_2, x_3, \dots, x_n$ are variables

b is a scalar value

For example:

Variables or Unknowns

$$3x_1 - 5x_2 + x_3 = 2$$

$a_1 \rightarrow 3$ $a_2 \rightarrow -5$ $a_3 \rightarrow 1$ $b \rightarrow 2$

Linear Equation

$$3x_1 - 5x_2 + x_3 = \sqrt{2} \quad \leftarrow \text{Linear Equation}$$

$$3x_1 - \sqrt{5}x_2 + x_3 = 2 \quad \leftarrow \text{Linear Equation}$$

$$3x_1 - 5\sqrt{x_2} + x_3 = 2 \quad \leftarrow \text{Non-Linear Equation}$$

$$3x_1 - 5x_2x_3 = 2 \quad \leftarrow \text{Non-Linear Equation}$$

System of Linear Equations

- System of Linear equations is a collection of one or more linear equations involving same variables.

$$\begin{array}{rcccl} 3x_1 & - & 5x_2 & + & x_3 & = & 2 \\ \updownarrow & & \updownarrow & & \updownarrow & & \\ -7x_1 & + & x_2 & + & 3x_3 & = & -1 \end{array}$$

Linear System of 3 Unknowns

$$\begin{array}{rcccl} 2x_1 & - & 5x_2 & = & 3 \\ \updownarrow & & \updownarrow & & \\ -x_1 & + & 3x_2 & = & -1 \end{array}$$

Linear System of 2 Unknowns

System of Linear Equations

- System of linear equations is a collection of one or more linear equations involving same variables.

$$a_1x_1 + a_2x_2 + \cdots \dots + a_nx_n = b$$

Linear Equation of n Unknowns

Solution

$$x_1 \rightarrow ? \quad x_2 \rightarrow ? \quad x_3 \rightarrow ? \quad x_n \rightarrow ?$$

Satisfy the Linear Equations

System of Linear Equations

Example

$$2x_1 - 5x_2 = 3$$

$$-x_1 + 3x_2 = -1$$

Linear System of 2 Unknowns: x_1, x_2

$$x_1 = 4 \quad x_2 = 1$$

$$2x_1 - 5x_2 = 3$$

$$2 * 4 - 5 * 1 = 3$$

$$8 - 5 = 3$$

$$3 = 3$$

$$-x_1 + 3x_2 = -1$$

$$-1 * 4 + 3 * 1 = -1$$

$$-4 + 3 = -1$$

$$-1 = -1$$

System of Linear Equations

- A system of linear equations is a collection of one or more linear equations involving same variables.
- A system of linear equations may have:
 - ✓ exactly **one solution**
 - ✓ infinitely **many solutions**
 - ✓ **no solution**

System of Linear Equations

- Exactly one solution

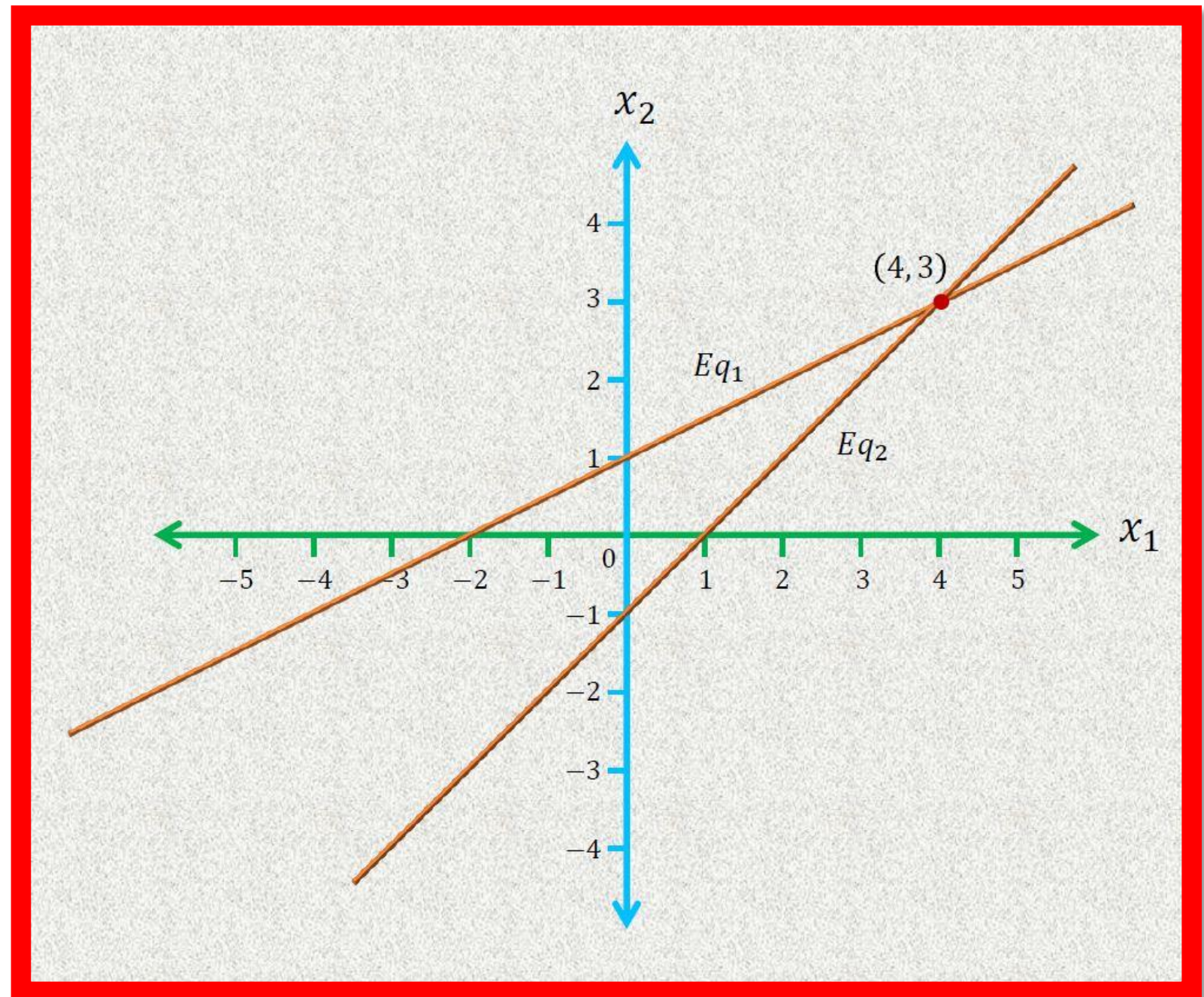
$$x_1 - 2x_2 = -2$$

$$x_1 - x_2 = 1$$

Solution:

$$x_1 = 4$$

$$x_2 = 3$$



System of Linear Equations

- Infinitely many solutions

$$x_1 - 2x_2 = -2$$

$$-x_1 + 2x_2 = 2$$

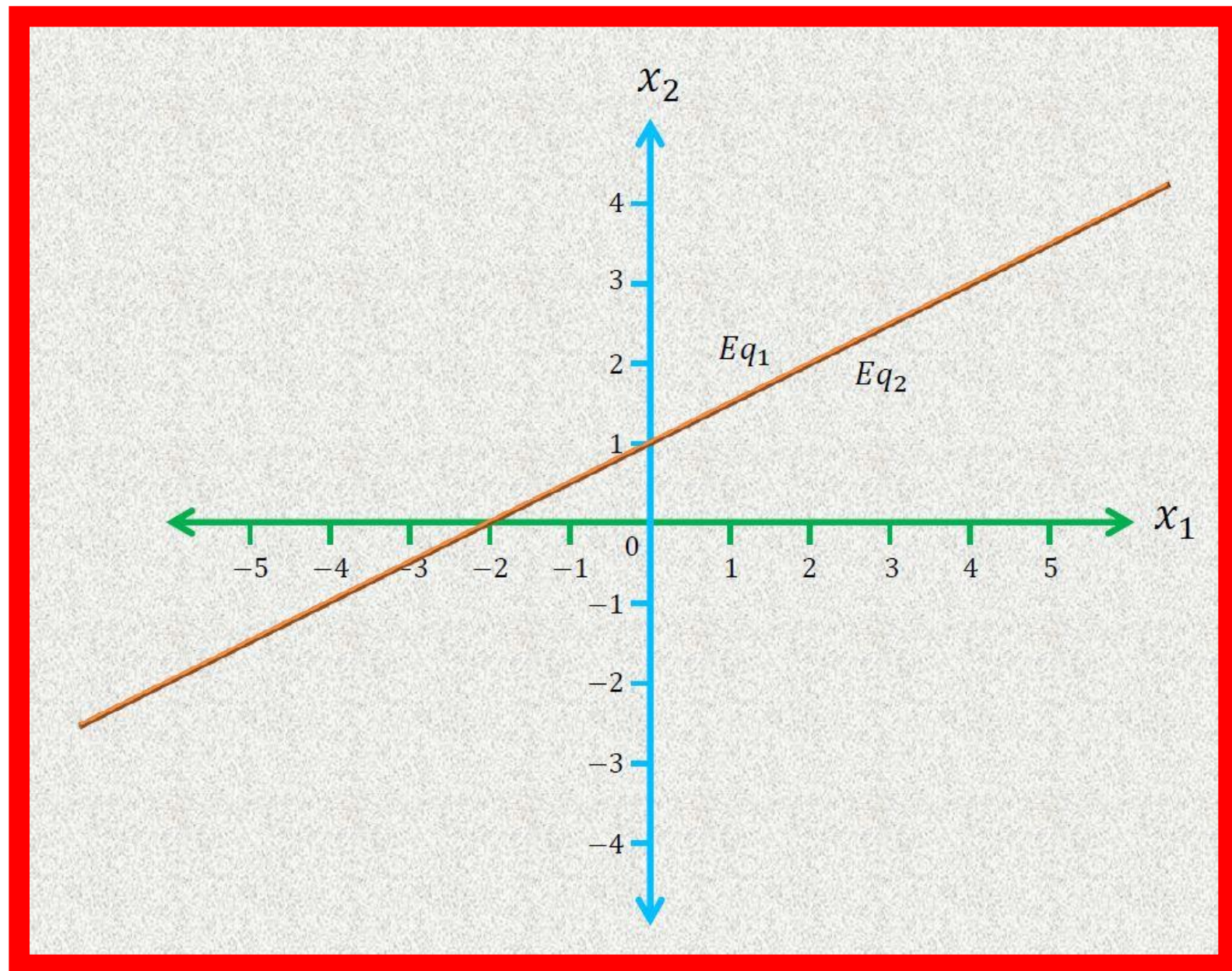
Solution:

$$x_1 =$$

$$x_2 =$$



*Infinite
Solutions*



System of Linear Equations

- No solution

$$x_1 - 2x_2 = -2$$

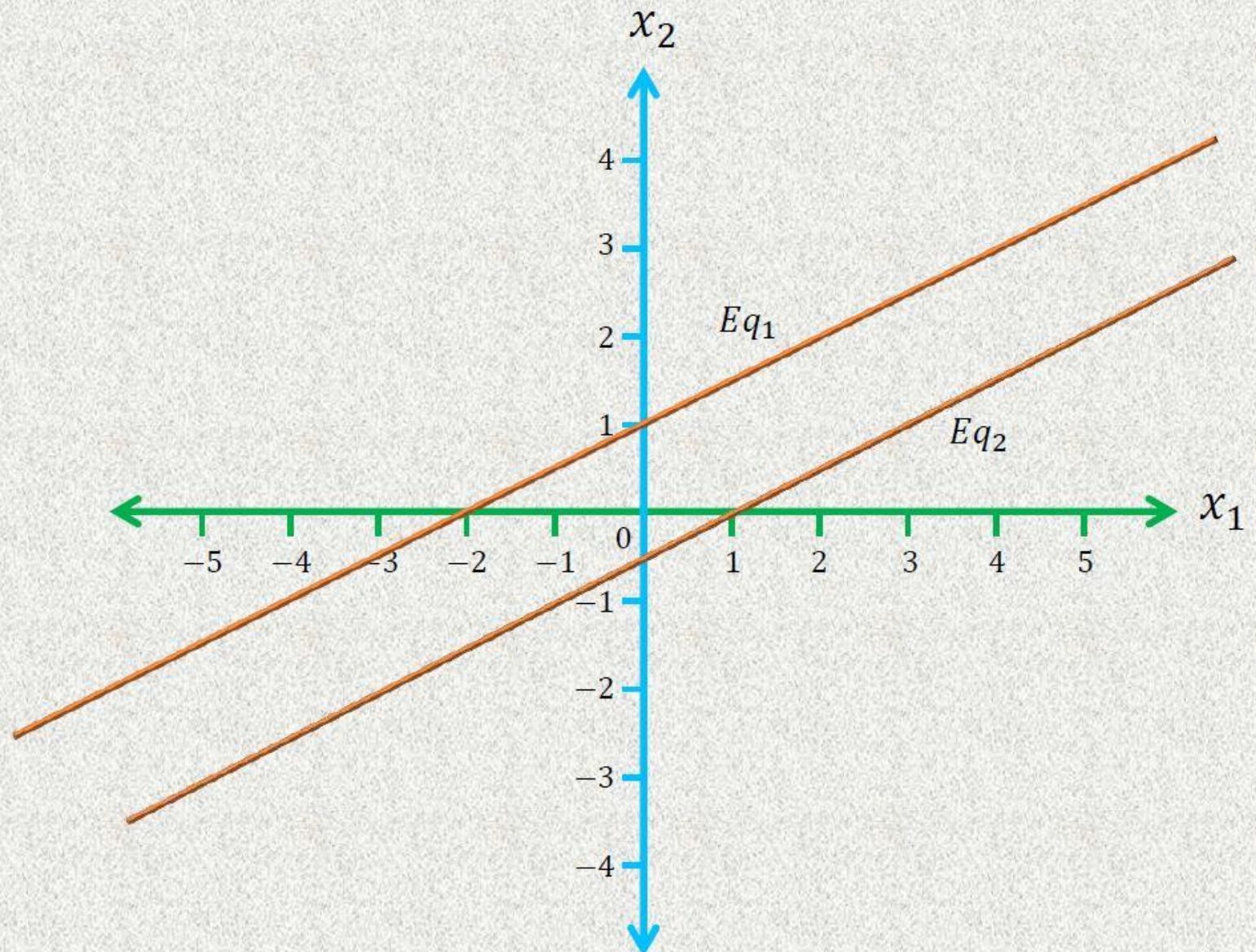
$$x_1 - 2x_2 = 1$$

Solution:

$$x_1 =$$

$$x_2 =$$

*No
Solution*



System of Linear Equations

- A system of linear equations is a collection of one or more linear equations involving same variables.
- System of linear equations may have:
 - ✓ exactly one solution -----> consistent system
 - ✓ infinitely many solutions -----> consistent system
 - ✓ no solution -----> Inconsistent system

Matrix Notation

- We can represent a system of linear equations using **matrices**.

$$\begin{aligned}x_1 - 2x_2 &= -2 \\x_1 - x_2 &= 1\end{aligned}$$

$$\left[\begin{array}{cc|c} 1 & -2 & -2 \\ 1 & -1 & 1 \end{array} \right]$$

Augmented Matrix

$$\begin{aligned}x_1 - x_2 + 2x_3 &= -4 \\x_2 - 3x_3 &= 9 \\2x_1 + x_3 &= 1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

Elementary Row Operations

- **Elementary row operations** are fundamental tools used in matrix algebra to solve systems of linear equations.
 - Interchange operation (row switching)
 - Scaling operation (row multiplication)
 - Replacement operation (row addition)
- Elementary row operations are designed to transform a matrix in ways that preserve the solution set of the associated system of linear equations.
- The operations lead to a new matrix that represents the same system of linear equations as the original matrix. Therefore, the solutions to the system are **not affected** by applying elementary row operations.

Elementary Row Operations

Interchange operation

$$\begin{aligned}x_1 - x_2 + 2x_3 &= -4 \\x_2 - 3x_3 &= 9 \\2x_1 \quad \quad + x_3 &= 1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

$R_2 \leftrightarrow R_3$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$



$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 2 & 0 & 1 & 1 \\ 0 & 1 & -3 & 9 \end{array} \right]$$

Elementary Row Operations

Scaling operation

$$\begin{aligned}x_1 - x_2 + 2x_3 &= -4 \\x_2 - 3x_3 &= 9 \\2x_1 \quad \quad + x_3 &= 1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$



$$\underline{R_1 \leftarrow 2R_1}$$

$$\left[\begin{array}{ccc|c} 2 & -2 & 4 & -8 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Elementary Row Operations

Replacement operation

$$\begin{aligned}x_1 - x_2 + 2x_3 &= -4 \\x_2 - 3x_3 &= 9 \\2x_1 \quad \quad + x_3 &= 1\end{aligned}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$



$$\underline{R_3 \leftarrow R_3 - 2R_1}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 2 & -3 & 9 \end{array} \right]$$

Gaussian Elimination Method

- When a system of linear equations is represented using an **augmented matrix**, **elementary row operations** can be performed to solve the system.
- **Gaussian elimination method** is a technique to change an augmented matrix into an **upper triangular matrix**.
- **Gaussian elimination method** turns augmented matrix into **row-echelon** form.

Gaussian Elimination Method

- A matrix is in **row-echelon** form if it satisfies the following conditions:
 - ✓ **Leading Entries**: In each row, the first non-zero entry (from the left) is called the **leading entry** or **leading coefficient**. The leading entry must be **1**.
 - ✓ **Zero Rows**: Any row consisting entirely of zeros must be at the bottom of the matrix.
 - ✓ **Staircase Pattern**: The leading entry of each non-zero row must be to the right of the leading entry of the row above it. This creates a **staircase** of leading **1**s.
 - ✓ **Zeros Below Leading Entries**: The column containing the leading entry of a row has zeros in all positions below this leading entry.
- **Row-echelon form** is particularly useful in solving systems of linear equations using **Gaussian elimination**, as it simplifies back-substitution.

Gaussian Elimination Method

Example of a Matrix in Row-Echelon Form

$$\begin{pmatrix} 1 & a & b & c \\ 0 & 1 & d & e \\ 0 & 0 & 1 & f \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Gaussian Elimination Method

Example

$$x_1 - x_2 + 2x_3 = -4$$

$$x_2 - 3x_3 = 9$$

$$2x_1 + x_3 = 1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

Gaussian Elimination Method

Example

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

$$\underline{R_3 \leftarrow R_3 - 2R_1}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 2 & -3 & 9 \end{array} \right]$$



$$\underline{R_3 \leftarrow R_3 - 2R_2}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 3 & -9 \end{array} \right]$$



$$\underline{R_3 \leftarrow \frac{1}{3}R_3}$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

Gaussian Elimination Method

Example

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right] \xrightarrow[\text{Elimination Method}]{\text{Gauss}} \left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\underline{x_3 = -3}$$

$$x_2 - 3x_3 = 9$$

$$x_1 - x_2 + 2x_3 = -4$$

$$x_2 - 3 * -3 = 9$$

$$x_1 - 0 + 2 * -3 = -4$$

$$x_2 + 9 = 9$$

$$x_1 - 6 = -4$$

$$\underline{x_2 = 0}$$

$$\underline{x_1 = 2}$$

$$x_1 = 2$$

$$x_2 = 0$$

$$x_3 = -3$$

Gauss-Jordan Elimination Method

- **Gauss-Jordan elimination** transforms a matrix into reduced row-echelon form through a series of row operations, simplifying the process of solving linear equations and determining matrix properties.
- A matrix is in reduced row-echelon form if it satisfies the following conditions:
 - ✓ **Leading entry is 1**: Each row that contains a non-zero element has a **leading entry** (the first non-zero entry from the left) of **1**.
 - ✓ **Leading 1's are the Only Non-Zero Entries in their Columns**: Each column containing a leading **1** has all other entries as zero.
 - ✓ **Leading 1's Move Rightward**: The leading **1** in any row is to the right of the leading **1** in the row above it.
 - ✓ **Rows at the Bottom**: Any row that contains only zeroes is at the bottom of the matrix.

Gauss-Jordan Elimination Method

Example of a Matrix in Reduced Row-Echelon Form (RREF)

$$\begin{pmatrix} 1 & 0 & 0 & 0 & a \\ 0 & 1 & 0 & 0 & b \\ 0 & 0 & 1 & 0 & c \\ 0 & 0 & 0 & 1 & d \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Gauss-Jordan Elimination Method

Example

$$x_1 - x_2 + 2x_3 = -4$$

$$x_2 - 3x_3 = 9$$

$$2x_1 + x_3 = 1$$

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right]$$

Augmented Matrix

Gauss-Jordan Elimination Method

Example

Augmented Matrix:

$$\left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 2 & 0 & 1 & 1 \end{array} \right] \xrightarrow[\text{Elimination Method}]{\text{Gauss}} \left[\begin{array}{ccc|c} 1 & -1 & 2 & -4 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\underline{R_1 \leftarrow R_1 + R_2}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & -1 & 5 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\underline{R_1 \leftarrow R_1 + R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & -3 & 9 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\underline{R_2 \leftarrow R_2 + 3R_3}$$

$$\left[\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & -3 \end{array} \right]$$

$$\begin{array}{l} x_1 = 2 \\ x_2 = 0 \\ x_3 = -3 \end{array}$$

Example of Transforming a Matrix to RREF

$$\begin{pmatrix} 2 & 1 & -1 & -1 \\ -3 & -1 & 2 & 3 \\ -2 & 1 & 2 & 2 \end{pmatrix}$$

Divide the first row by 2 to make the leading entry 1:

$$R1 \rightarrow \frac{1}{2}R1$$

$$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ -3 & -1 & 2 & 3 \\ -2 & 1 & 2 & 2 \end{pmatrix}$$

Example of Transforming a Matrix to RREF

Eliminate entries below the leading 1 in the first column:

$$R2 \rightarrow R2 + 3R1 \quad \text{and} \quad R3 \rightarrow R3 + 2R1$$

$$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & \frac{1}{2} & \frac{1}{2} & \frac{3}{2} \\ 0 & 2 & 1 & 1 \end{pmatrix}$$

Multiply the second row by 2 to make the leading entry 1:

$$R2 \rightarrow 2R2$$

$$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 1 & 3 \\ 0 & 2 & 1 & 1 \end{pmatrix}$$

Example of Transforming a Matrix to RREF

Eliminate the entry below the leading 1 in the second column:

$$R3 \rightarrow R3 - 2R2$$

$$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & -5 \end{pmatrix}$$

Multiply the third row by -1 to make the leading entry 1:

$$R3 \rightarrow -1R3$$

$$\begin{pmatrix} 1 & \frac{1}{2} & -\frac{1}{2} & -\frac{1}{2} \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & 5 \end{pmatrix}$$

Example of Transforming a Matrix to RREF

Eliminate the entries above the leading 1 in the third column:

$$R2 \rightarrow R2 - R3 \quad \text{and} \quad R1 \rightarrow R1 + \frac{1}{2}R3$$

$$\begin{pmatrix} 1 & \frac{1}{2} & 0 & 2 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \end{pmatrix}$$

Eliminate the entry above the leading 1 in the second column:

$$R1 \rightarrow R1 - \frac{1}{2}R2$$

$$\begin{pmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 5 \end{pmatrix}$$

Exercise

- Solve the following linear system by both Gaussian elimination and Gauss-Jordan elimination methods.

$$\begin{array}{rcl} x + 2y + 3z & = & 6 \\ 2x - 3y + 2z & = & 14 \\ 3x + y - z & = & -2. \end{array}$$

End of Lecture 4



MATHEMATICS FOR MACHINE LEARNING

Marc Peter Deisenroth
A. Aldo Faisal